
RESEARCH PROJECT REPORT

Distribution-Aware PCA

Geometry and Optimization of Information-Theoretic Projection Objectives

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A generalized framework for dimensionality reduction in which the projection matrix is obtained by maximizing a distribution-specific information-theoretic objective on the Stiefel/Grassmann manifold

Central object	Linear projection $Y = W^\top X$, $W^\top W = I_k$
Governing problem	$W^* = \arg \max_{W^\top W = I_k} J(W)$
Cases studied	Gaussian, Student- t , Laplace (ℓ_1)
Manifolds	Stiefel $\text{St}(d, k)$, Grassmann $\text{Gr}(k, d)$

Abstract

Classical Principal Component Analysis (PCA) selects projection directions that maximize retained variance, equivalently minimizing mean-squared reconstruction error. This is only one possible definition of “retained information.” This report develops a general framework,

$$W^* = \arg \max_{W^\top W = I_k} J(W), \quad W \in \mathbb{R}^{d \times k},$$

in which the objective $J(W)$ is induced by an explicit probabilistic model of the data, rather than fixed in advance to be the variance. We show, with complete derivations, that under a Gaussian model the information-theoretic (entropy) objective and the classical variance objective are distinct functions of W that nevertheless share the same maximizer — the top- k eigenspace of the covariance matrix — which is the precise mathematical sense in which “PCA is the Gaussian special case” of a broader family. We then examine two non-Gaussian cases, Student- t and Laplace, which change the *geometry* of the optimization problem: the Student- t objective remains smooth but becomes non-convex with no closed form, while the Laplace objective becomes non-smooth. For each case we derive the objective from its likelihood, compute gradients (or subgradients) explicitly, characterize invariance under the constraint manifold, and connect these geometric properties to a specific, justified choice of numerical optimization algorithm on the Stiefel and Grassmann manifolds. The report closes with a full experimental protocol, an implementation roadmap, and a consolidated reference table of every objective, gradient, and recommended solver derived in the text.

Keywords: Principal Component Analysis, differential entropy, mutual information, maximum likelihood, Stiefel manifold, Grassmann manifold, Riemannian optimization, robust statistics, heavy-tailed distributions, non-smooth optimization.

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CHAPTER 1 Introduction and Motivation

1.1 From Variance to Information

Principal Component Analysis reduces a d -dimensional random vector X to a k -dimensional vector $Y = W^\top X$ ($k < d$) by choosing the orthonormal matrix W that keeps as much *variance* as possible. Variance, however, is only a second-moment summary of a distribution: it is the natural notion of “information” for Gaussian data, but it is not the only possible notion, and it is not obviously the correct one once the data-generating distribution is no longer Gaussian.

This report is organized around one guiding question, stated formally:

Guiding question. For a chosen probability distribution and a chosen notion of statistical information, what objective function $J(W)$ should be optimized over projection matrices W , and what is the mathematical geometry — continuity, smoothness, invariance, convexity, and landscape — of that objective?

1.2 Formal Statement of the Project

The unifying optimization problem studied throughout this report is

$$W^\star = \arg \max_{W^\top W = I_k} J(W) \tag{1.1}$$

where:

- $W \in \mathbb{R}^{d \times k}$ is the projection matrix,
- the constraint $W^\top W = I_k$ forces the k columns of W to be orthonormal, which is exactly the condition that makes $Y = W^\top X$ a well-posed, non-degenerate, rotation-free coordinate system for the reduced representation,
- $J(W)$ is a *distribution-aware information objective*: entropy, likelihood, mutual information, or a robust surrogate built from an assumed density for Y .

The project decomposes into four linked tasks, executed for three representative distributions (Gaussian, Student- t , Laplace):

1. **Derive** $J(W)$ explicitly from the assumed distribution.
2. **Analyze** the geometry of J on the constraint set: is it continuous, differentiable, basis-invariant, convex/non-convex, and what does its critical-point structure look like?
3. **Select** a numerical optimization method whose applicability is justified *by* that geometry, rather than chosen by default.
4. **Compare** the resulting projections against classical PCA on synthetic data with known ground truth.

1.3 Why This Is a Genuine Generalization, Not a Relabeling

It would be easy to simply rename PCA’s variance objective as an “information” objective without changing any mathematics. The content of this report is to show, by explicit computation, that:

- entropy maximization under a Gaussian model produces a *different function of W* than the variance objective (Chapter 6) — a log-determinant instead of a trace — and yet the two share the same maximizer, which is a non-trivial fact requiring proof, not an assumption;
- entropy/likelihood maximization under non-Gaussian models produces objectives with *qualitatively different geometry* (smooth but non-convex for Student- t ; non-smooth for Laplace), which forces a change of numerical algorithm — this is the real payoff of framing PCA information-theoretically.

1.4 Roadmap

Chapter 2 builds the necessary information theory from first principles (surprisal, entropy, differential entropy, mutual information, and the Gaussian maximum-entropy property). Chapter 3 poses dimensionality reduction as a constrained information-maximization problem. Chapter 4 re-derives classical PCA in full detail from three equivalent viewpoints. Chapter 5 introduces the entropy-based generalization. Chapter 6 works out the Gaussian case completely, including every gradient. Chapter 7 develops the Stiefel/Grassmann manifold machinery needed for all subsequent optimization. Chapter 8 derives the Student- t and Laplace objectives. Chapter 9 gives the general geometric-analysis checklist. Chapter 10 presents the optimization algorithms implied by that geometry. Chapter 11 consolidates the method-selection logic into a decision table. Chapters 12–13 give the experimental protocol and implementation roadmap. Chapter 14 concludes.

CHAPTER 2 Foundations of Information Theory

This chapter builds, from first principles, every information-theoretic quantity used later: surprisal, entropy, differential entropy, and mutual information. Each definition is followed by the properties that matter for the rest of the report, proved in full.

2.1 Surprisal: The Information Content of a Single Outcome

Let X be a discrete random variable taking value x with probability $p(x)$. We want a function $I(x)$ quantifying “how much information is revealed by observing $X = x$,” subject to three intuitive requirements:

1. $I(x)$ should depend only on $p(x)$, decreasingly: rarer events are more informative, so I should be a decreasing function of p .
2. Certain events carry no information: $p(x) = 1 \Rightarrow I(x) = 0$.
3. Information from independent events should add: if X, Z are independent, observing both should give $I(x, z) = I(x) + I(z)$, while $p(x, z) = p(x)p(z)$.

Requirement 3 demands a function f with $f(p(x)p(z)) = f(p(x)) + f(p(z))$ for all valid probabilities — a Cauchy-type functional equation. Together with monotonicity and $f(1) = 0$, the unique (continuous) solution is $f(p) = -c \log p$ for a positive constant c , which fixes the base of the logarithm (a choice of unit) but not the functional form. Setting $c = 1$ (natural units, “nats”) gives

$$I(x) = -\log p(x) \tag{2.1}$$

which satisfies all three requirements: it is decreasing in $p(x)$ because $-\log$ is decreasing; $-\log 1 = 0$; and $-\log(p(x)p(z)) = -\log p(x) - \log p(z) = I(x) + I(z)$.

2.2 Shannon Entropy: Average Information

Definition 2.1 (Entropy). The entropy of a discrete random variable X is the expected surprisal,

$$H(X) = \mathbb{E}[-\log p(X)] = -\sum_x p(x) \log p(x). \tag{2.2}$$

Bounds on entropy — full derivation

For X taking n possible values, $0 \leq H(X) \leq \log n$.

Lower bound. Each term $-p(x) \log p(x) \geq 0$ because $0 \leq p(x) \leq 1 \Rightarrow \log p(x) \leq 0$. Hence $H(X) \geq 0$, with equality iff every term vanishes, i.e. each $p(x) \in \{0, 1\}$ — the deterministic case:

$$P(X = a) = 1 \implies H(X) = -1 \cdot \log 1 = 0.$$

Upper bound (via Jensen's inequality). Since $\log$ is concave,

$$\begin{aligned} H(X) &= \sum_x p(x) \log \frac{1}{p(x)} \\ &\leq \log \left(\sum_x p(x) \cdot \frac{1}{p(x)} \right) \quad (\text{Jensen, concave } \log) \\ &= \log \left(\sum_{x: p(x) > 0} 1 \right) = \log n. \end{aligned} \tag{2.3}$$

Equality in Jensen holds iff $1/p(x)$ is constant over the support, i.e. the uniform distribution $p(x) = 1/n$. **Example (fair coin):** $P(H) = P(T) = \frac{1}{2}$ gives $H(X) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = \log 2$, which in base-2 units is exactly 1 bit — the maximum possible for a binary variable, consistent with (2.3) at $n = 2$.

2.3 Differential Entropy

PCA acts on continuous vectors, so we need the continuous analogue.

Definition 2.2 (Differential entropy). If $X \in \mathbb{R}^d$ has density $p(x)$,

$$h(X) = - \int_{\mathbb{R}^d} p(x) \log p(x) dx = \mathbb{E}[-\log p(X)]. \tag{2.4}$$

Two structural differences from discrete entropy

(i) **Differential entropy can be negative.** Take $X \sim \text{Unif}(0, a)$ with $a < 1$. Then $p(x) = 1/a$ on $(0, a)$, and

$$h(X) = - \int_0^a \frac{1}{a} \log \frac{1}{a} dx = -\log \frac{1}{a} = \log a < 0 \quad \text{for } a < 1.$$

This is impossible for discrete entropy, which is always ≥ 0 .

(ii) **Differential entropy is not invariant under general coordinate changes.** Under an invertible linear map $X' = AX$ ($A \in \mathbb{R}^{d \times d}$), the change-of-variables formula gives density $p_{X'}(x') = p_X(A^{-1}x')/|\det A|$, and a direct computation gives

$$h(AX) = h(X) + \log |\det A|. \tag{2.5}$$

Derivation:

$$\begin{aligned} h(AX) &= - \int p_{X'}(x') \log p_{X'}(x') dx' \\ &= - \int \frac{p_X(A^{-1}x')}{|\det A|} \log \frac{p_X(A^{-1}x')}{|\det A|} dx' \end{aligned}$$

substitute $x = A^{-1}x'$, $dx' = |\det A| dx$:

$$\begin{aligned} &= - \int p_X(x) \left[\log p_X(x) - \log |\det A| \right] dx \\ &= h(X) + \log |\det A|. \end{aligned}$$

So scaling coordinates changes differential entropy by an additive constant — it is a valid *relative* measure of spread, but the absolute number depends on the coordinate system, unlike discrete entropy. This subtlety is why entropy-based projection objectives must be compared under a fixed orthonormal constraint ($\det A = \pm 1$ for orthonormal A , so $\log |\det A| = 0$ and entropy differences between orthonormal projections are coordinate-free).

2.4 Mutual Information

Definition 2.3 (Mutual information).

$$I(X; Y) = H(X) - H(X | Y) = h(X) - h(X | Y) \quad (2.6)$$

for the continuous case, with the equivalent symmetric form

$$I(X; Y) = h(X) + h(Y) - h(X, Y). \quad (2.7)$$

Non-negativity of mutual information — full proof

Proposition 2.4. $I(X; Y) \geq 0$, with equality iff $X \perp Y$.

Proof. Write $I(X; Y) = \mathbb{E} \left[\log \frac{p(X, Y)}{p(X)p(Y)} \right]$, which follows directly from (2.7) by expanding each entropy as an expected log-density. By Jensen's inequality applied to the concave function $\log$,

$$\begin{aligned} I(X; Y) &= -\mathbb{E} \left[\log \frac{p(X)p(Y)}{p(X, Y)} \right] \\ &\geq -\log \mathbb{E} \left[\frac{p(X)p(Y)}{p(X, Y)} \right] \\ &= -\log \int \int p(x)p(y) dx dy = -\log 1 = 0. \end{aligned}$$

Equality in Jensen requires the argument $\frac{p(X)p(Y)}{p(X, Y)}$ to be almost-surely constant, and since its expectation is 1, the constant must be 1, i.e. $p(X, Y) = p(X)p(Y)$ a.e. — exactly independence. $\square$

Interpretation. $I(X; Y)$ measures how much observing Y reduces uncertainty about X ; it is symmetric, non-negative, and zero exactly under independence. This motivates the mutual-information projection objective of §3.2.3, though it also explains its pathology: for a noiseless deterministic map $Y = W^\top X$, one can push $I(X; Y) \rightarrow \infty$ by simply not reducing dimension, so mutual information alone is not directly usable without regularization.

2.5 Maximum Entropy Property of the Gaussian

The reason the Gaussian model reproduces PCA (Chapter 6) is a special case of a deeper fact, proved here.

Theorem 2.5 (Gaussian maximizes differential entropy at fixed covariance). *Among all densities p on $\mathbb{R}^k$ with mean μ and covariance Σ , the entropy $h(p)$ is maximized uniquely by $p = \mathcal{N}(\mu, \Sigma)$.*

Proof. Let $q = \mathcal{N}(\mu, \Sigma)$ and let p be any competitor density with the same mean and covariance. Non-negativity of Kullback–Leibler divergence gives $D(p||q) = \int p \log(p/q) \geq 0$, i.e. $\int p \log p \geq \int p \log q$. Because $\log q(x)$ is a *quadratic* function of x (from the Gaussian exponent) and p shares its first and second moments with q ,

$$\int p(x) \log q(x) dx = \int q(x) \log q(x) dx,$$

since both integrals reduce to the same polynomial functional of μ, Σ regardless of the higher-order shape of p . Combining,

$$-h(p) = \int p \log p \geq \int p \log q = \int q \log q = -h(q),$$

so $h(p) \leq h(q)$, i.e. the Gaussian has maximal entropy for the given covariance, with equality iff $D(p||q) = 0$, i.e. $p = q$. $\square$

This theorem is the deep reason that *assuming Gaussianity and maximizing entropy* is equivalent to *using only second-moment (covariance) information* — which is precisely what classical PCA does. Non-Gaussian objectives (Chapter 8) depart from this picture by using higher-order distributional information.

CHAPTER 3 — Dimensionality Reduction as an Information-Theoretic Problem

3.1 The Linear Projection Model

Let $X \in \mathbb{R}^d$ be the (mean-zero, without loss of generality) data vector. A k -dimensional linear representation is

$$Y = W^\top X, \quad W \in \mathbb{R}^{d \times k}, \quad W^\top W = I_k. \quad (3.1)$$

The orthonormality constraint is not cosmetic: without it, the objective could be inflated arbitrarily by rescaling columns of W (e.g. any variance-type objective grows without bound as $\|w\| \rightarrow \infty$), and the columns would no longer represent genuinely distinct, non-redundant directions. Constraining W to the **Stiefel manifold** $\text{St}(d, k) = \{W \in \mathbb{R}^{d \times k} : W^\top W = I_k\}$ is what makes the problem well-posed.

3.2 Five Candidate Notions of “Preserved Information”

1. **Variance / reconstruction** — classical PCA (Chapter 4).
2. **Differential entropy of the projection**, $J(W) = h(W^\top X)$ (Chapter 5 onward).
3. **Mutual information** between original and reduced data, $J(W) = I(X; W^\top X)$ (§3.2.3).
4. **Likelihood** under an assumed parametric model for the projected coordinates, $J(W) = \sum_i \log p(W^\top x_i; \theta(W))$.
5. **Robust scatter** objectives derived from heavy-tailed likelihoods (Chapter 8).

3.2.1 Option A: Projected differential entropy

$$J(W) = h(W^\top X). \quad (3.2)$$

Larger entropy means the projected variable is less predictable, less collapsed, and retains more of the “spread” of the original distribution. A projection that collapses X onto a near-degenerate subspace produces a low-entropy, highly concentrated Y .

3.2.2 Option B: Projected log-likelihood

If $Y = W^\top X$ is modeled as following density $p(y; \theta(W))$,

$$J(W) = \sum_{i=1}^n \log p(W^\top x_i; \theta(W)). \quad (3.3)$$

This is the maximum-likelihood viewpoint: choose W (and possibly model parameters θ jointly) so the projected sample looks as likely as possible under the assumed family. For an exponential-family model this often reduces to, or closely tracks, the entropy objective.

3.2.3 Option C: Mutual information

$$J(W) = I(X; W^\top X). \quad (3.4)$$

By (2.6), for a deterministic noiseless linear map, $I(X; W^\top X) = h(X) - h(X \mid W^\top X)$, and since X is a deterministic function of $(W^\top X, W_\perp^\top X)$ for any completion $W_\perp$, this quantity can diverge or behave degenerately (it does not naturally penalize dimension reduction the way one intuitively wants). In practice (3.4) is used through a noisy-channel surrogate ($Y = W^\top X + \varepsilon$) or a model-based approximation, rather than directly.

3.2.4 Option D: Robust scatter

If X is assumed heavy-tailed, the maximum-likelihood objective naturally becomes a *robust* scatter criterion — large deviations are penalized sub-quadratically rather than quadratically. This is worked out fully in Chapter 8.

3.3 The General Optimization Problem, Restated

Collecting the discussion, the master problem of this report is

$$\boxed{W^\star = \arg \max_{W \in \text{St}(d,k)} J(W)} \quad (3.5)$$

and the project's real content is: (a) computing $J(W)$ explicitly for each distributional assumption, and (b) determining what optimization algorithm the resulting geometry of J requires — the subject of Chapters 6–10.

CHAPTER 4 ————— Classical PCA: A Complete Derivation

Before generalizing, we re-derive PCA itself with full rigor, since every later comparison (“does entropy give the same answer as variance?”) requires an exact statement of the classical baseline.

4.1 The One-Dimensional Problem via Lagrange Multipliers

Let $X \in \mathbb{R}^d$ have mean 0 and covariance $\Sigma \succeq 0$. For a unit vector $w \in \mathbb{R}^d$, the variance of the projection $y = w^\top X$ is

$$\text{Var}(w^\top X) = \mathbb{E}[(w^\top X)^2] = w^\top \mathbb{E}[XX^\top]w = w^\top \Sigma w.$$

The first principal direction solves

$$\max_{\|w\|=1} w^\top \Sigma w. \quad (4.1)$$

Full Lagrangian solution. Introduce a multiplier λ for the constraint $w^\top w = 1$ and form the Lagrangian

$$\mathcal{L}(w, \lambda) = w^\top \Sigma w - \lambda(w^\top w - 1).$$

Stationarity requires $\nabla_w \mathcal{L} = 0$:

$$\nabla_w \mathcal{L} = 2\Sigma w - 2\lambda w = 0 \implies \Sigma w = \lambda w.$$

So every stationary point w is an eigenvector of Σ , with λ the corresponding eigenvalue. Evaluating the objective at such a point:

$$w^\top \Sigma w = w^\top (\lambda w) = \lambda w^\top w = \lambda.$$

Hence the objective value at a stationary point equals its eigenvalue, so the *maximum* over all unit eigenvectors is attained at the eigenvector for the **largest** eigenvalue λ_1 .

Proposition 4.1 (Rayleigh quotient bound). *For any $\Sigma = \Sigma^\top$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$, and any unit vector w , $\lambda_d \leq w^\top \Sigma w \leq \lambda_1$, with the upper (resp. lower) bound attained exactly at the top (resp. bottom) eigenvector.*

Proof. Diagonalize $\Sigma = U\Lambda U^\top$ with U orthogonal, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$. Let $v = U^\top w$, so $\|v\| = \|w\| = 1$. Then

$$w^\top \Sigma w = v^\top \Lambda v = \sum_{i=1}^d \lambda_i v_i^2 \leq \lambda_1 \sum_i v_i^2 = \lambda_1,$$

since $\lambda_i \leq \lambda_1$ for all i and $\sum v_i^2 = 1$. The lower bound is symmetric. Equality holds iff all mass of v sits on the top (bottom) eigenspace. $\square$

4.2 The k -Dimensional Problem: Trace Maximization

For k components, $W = [w_1, \dots, w_k] \in \mathbb{R}^{d \times k}$, $W^\top W = I_k$, the total retained variance is

$$\sum_{j=1}^k \text{Var}(w_j^\top X) = \sum_{j=1}^k w_j^\top \Sigma w_j = \text{tr}(W^\top \Sigma W).$$

So PCA solves

$$\boxed{\max_{W^\top W = I_k} \text{tr}(W^\top \Sigma W).} \quad (4.2)$$

Theorem 4.2 (Ky Fan). *The maximum of (4.2) equals $\lambda_1 + \dots + \lambda_k$ (the sum of the k largest eigenvalues of Σ), attained whenever the columns of W span the top- k eigenspace of Σ .*

Proof. Diagonalize $\Sigma = U \Lambda U^\top$ as above and set $V = U^\top W \in \mathbb{R}^{d \times k}$, which also satisfies $V^\top V = W^\top U U^\top W = W^\top W = I_k$ (columns of V remain orthonormal). Then

$$\text{tr}(W^\top \Sigma W) = \text{tr}(V^\top \Lambda V) = \sum_{i=1}^d \lambda_i \sum_{j=1}^k V_{ij}^2 = \sum_{i=1}^d \lambda_i c_i,$$

where $c_i := \sum_{j=1}^k V_{ij}^2$ is the squared row-norm of V . Because V has orthonormal columns, $\sum_i c_i = \text{tr}(V^\top V) = k$ and, since V 's rows have norm ≤ 1 (as $V V^\top \preceq I_d$ when V has orthonormal columns, a consequence of $V^\top V = I_k$), each $c_i \in [0, 1]$. Maximizing $\sum_i \lambda_i c_i$ subject to $0 \leq c_i \leq 1$, $\sum_i c_i = k$ is a linear program whose optimum (by the rearrangement/majorization argument) places $c_i = 1$ on the k largest λ_i and $c_i = 0$ elsewhere, giving value $\lambda_1 + \dots + \lambda_k$. This is achieved exactly when V 's columns span the coordinate subspace of the top- k eigenvalues, i.e. when W 's columns span the top- k eigenspace of Σ . $\square$

4.3 Equivalence with Reconstruction-Error Minimization

Proposition 4.3.

$$\min_{W^\top W = I_k} \mathbb{E} \|X - W W^\top X\|^2 \iff \max_{W^\top W = I_k} \text{tr}(W^\top \Sigma W). \quad (4.3)$$

Proof. Expand the squared reconstruction error:

$$\begin{aligned} \mathbb{E} \|X - W W^\top X\|^2 &= \mathbb{E} [X^\top X] - 2 \mathbb{E} [X^\top W W^\top X] + \mathbb{E} [X^\top W W^\top W W^\top X] \\ &= \mathbb{E} \|X\|^2 - 2 \mathbb{E} [\text{tr}(W^\top X X^\top W)] + \mathbb{E} [\text{tr}(W^\top X X^\top W)] \end{aligned}$$

using $W^\top W = I_k$ so $W W^\top W W^\top = W W^\top$ and $X^\top W W^\top X = \text{tr}(W^\top X X^\top W)$:

$$= \mathbb{E} \|X\|^2 - \text{tr}(W^\top \Sigma W). \quad (4.4)$$

Since $\mathbb{E} \|X\|^2 = \text{tr}(\Sigma)$ is a fixed constant independent of W , minimizing the left side of (4.3) is exactly equivalent to maximizing $\text{tr}(W^\top \Sigma W)$, proving the claimed equivalence, and confirming both formulations select the same W . $\square$

4.4 Summary of Classical PCA Properties

Property	Value for $J_{\text{PCA}}(W) = \text{tr}(W^\top \Sigma W)$
Smoothness	C^∞ (quadratic polynomial in W)
Basis invariance	Yes: $J(WQ) = J(W)$ for all $Q \in O(k)$ (cyclic trace property)
Convexity in W (Euclidean)	Convex objective on a non-convex constraint set; no global convexity claim needed since a closed form exists
Closed-form solution	Yes — top- k eigenvectors of Σ
Manifold	Grassmann (basis-invariant, so only the subspace matters)

CHAPTER 5 ————— The Entropy-Maximization Generalization of PCA

5.1 Motivation

Classical PCA fixes, in advance, that “information” means “variance.” The entropy generalization instead asks: whatever the distribution of X happens to be, choose W so that the projected random vector $Y = W^\top X$ has the largest possible differential entropy. Formally,

$$\boxed{J(W) = h(W^\top X)}, \quad \max_{W^\top W = I_k} J(W). \quad (5.1)$$

5.2 Why This Is a Reasonable Generalization

- High entropy $\Leftrightarrow$ the projected representation is spread out and unpredictable; a degenerate or collapsed projection (e.g. onto a direction with near-zero variance, or onto a direction where the data pile up at a point) has low entropy.
- Because entropy is defined for *any* distribution, not just Gaussian, (5.1) is meaningful for every distributional assumption we might make about X — it is the correct common ancestor of all the case studies in this report.
- By §2.5, if X is (assumed) Gaussian, maximizing (5.1) uses only the covariance of X — precisely the ingredient classical PCA uses. If X is not Gaussian, (5.1) implicitly draws on higher-order information (skewness, kurtosis, tail weight) that variance discards.

This single objective, instantiated under three different distributional assumptions, generates the entire case-study structure of the remainder of the report.

CHAPTER 6 — The Gaussian Case in Full Mathematical Detail

6.1 Differential Entropy of a Multivariate Gaussian — Derived from Scratch

Let $Y \in \mathbb{R}^k$, $Y \sim \mathcal{N}(0, \Sigma_Y)$ with $\Sigma_Y \succ 0$. Its density is

$$p(y) = \frac{1}{(2\pi)^{k/2} |\Sigma_Y|^{1/2}} \exp\left(-\frac{1}{2} y^\top \Sigma_Y^{-1} y\right).$$

Step-by-step computation of $h(Y) = \mathbb{E}[-\log p(Y)]$. Take the log of the density:

$$\log p(y) = -\frac{k}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_Y| - \frac{1}{2} y^\top \Sigma_Y^{-1} y.$$

Negate and take expectation term by term:

$$h(Y) = \frac{k}{2} \log(2\pi) + \frac{1}{2} \log |\Sigma_Y| + \frac{1}{2} \mathbb{E}[Y^\top \Sigma_Y^{-1} Y].$$

The last term is evaluated using the trace trick: for any symmetric matrix A , $Y^\top A Y = \text{tr}(A Y Y^\top)$, so

$$\mathbb{E}[Y^\top \Sigma_Y^{-1} Y] = \text{tr}(\Sigma_Y^{-1} \mathbb{E}[Y Y^\top]) = \text{tr}(\Sigma_Y^{-1} \Sigma_Y) = \text{tr}(I_k) = k.$$

Substituting back,

$$h(Y) = \frac{k}{2} \log(2\pi) + \frac{1}{2} \log |\Sigma_Y| + \frac{k}{2} = \frac{1}{2} \left[k \log(2\pi) + k + \log |\Sigma_Y| \right] = \frac{1}{2} \log[(2\pi e)^k |\Sigma_Y|],$$

using $k \log(2\pi) + k = k \log(2\pi e)$ and $\log(ab) = \log a + \log b$.

$$\boxed{h(Y) = \frac{1}{2} \log[(2\pi e)^k \det \Sigma_Y]} \tag{6.1}$$

6.2 The Projected Gaussian Entropy Objective

If $X \sim \mathcal{N}(0, \Sigma)$, then $Y = W^\top X$ is Gaussian with $\Sigma_Y = W^\top \Sigma W$ (a direct consequence of linearity of Gaussians: $\mathbb{E}[Y Y^\top] = W^\top \mathbb{E}[X X^\top] W = W^\top \Sigma W$). By (6.1),

$$h(W^\top X) = \frac{1}{2} \log[(2\pi e)^k \det(W^\top \Sigma W)] = \underbrace{\frac{k}{2} \log(2\pi e)}_{\text{constant, independent of } W} + \frac{1}{2} \log \det(W^\top \Sigma W).$$

Dropping the additive constant (irrelevant to the location of the maximizer),

$$\boxed{J_G(W) = \log \det(W^\top \Sigma W)} \quad (6.2)$$

equivalently $\max_{W^\top W = I_k} \det(W^\top \Sigma W)$ since $\log$ is monotonic.

6.3 Comparing J_G and J_{PCA} : Same Optimizer, Different Function

Proposition 6.1. $J_G(W) = \log \det(W^\top \Sigma W)$ and $J_{\text{PCA}}(W) = \text{tr}(W^\top \Sigma W)$ are not the same function of W , yet both are maximized over $\text{St}(d, k)$ by exactly the same subspace: the top- k eigenspace of Σ .

Proof. Let $\Sigma = U \Lambda U^\top$, $\lambda_1 \geq \dots \geq \lambda_d$, and suppose W spans a k -dimensional coordinate subspace determined by index set $\{i_1, \dots, i_k\}$ (i.e. an invariant subspace of Σ , so that $W^\top \Sigma W$ is itself diagonal with entries $\lambda_{i_1}, \dots, \lambda_{i_k}$ — the natural candidates for extrema by the same symmetry argument as Theorem 4.2). Then

$$J_{\text{PCA}}(W) = \lambda_{i_1} + \dots + \lambda_{i_k}, \quad J_G(W) = \log(\lambda_{i_1} \lambda_{i_2} \dots \lambda_{i_k}) = \sum_{j=1}^k \log \lambda_{i_j}.$$

The functions $x \mapsto x$ and $x \mapsto \log x$ are both *strictly increasing*, so for any two eigenvalues $\lambda_a > \lambda_b$, swapping λ_b out for λ_a in the index set strictly increases *both* $\sum \lambda_{i_j}$ and $\sum \log \lambda_{i_j}$. Hence both sums, over all $\binom{d}{k}$ choices of k -subsets of eigenvalues, are maximized by the same choice: the k largest eigenvalues $\lambda_1, \dots, \lambda_k$. (A full optimization over all of $\text{St}(d, k)$, not just coordinate subspaces, is handled for J_G by the log-concavity / Hadamard-type argument below, confirming that non-invariant subspaces cannot do better.) Since sum and sum-of-logs are different functions ($x \mapsto x$ vs. $x \mapsto \log x$), $J_{\text{PCA}} \neq J_G$ as functions, yet $\arg \max J_{\text{PCA}} = \arg \max J_G$. $\square$

Remark 6.2. This is the precise, provable sense in which “PCA is the Gaussian entropy-maximizing projection.” It is not a definition or a relabeling — it is a theorem that two visibly different objective functions happen to share a maximizer, which follows from the eigenvalues entering both objectives only through order-preserving transformations (identity and $\log$, both strictly increasing).

6.4 Basis Invariance of J_G

Proposition 6.3. $J_G(WQ) = J_G(W)$ for every $Q \in O(k)$.

Proof.

$$J_G(WQ) = \log \det(Q^\top W^\top \Sigma W Q) = \log [\det(Q^\top) \det(W^\top \Sigma W) \det(Q)] = \log \det(W^\top \Sigma W) = J_G(W),$$

using multiplicativity of determinants and $\det(Q^\top) \det(Q) = \det(Q^\top Q) = \det(I_k) = 1$. $\square$

Hence J_G depends only on $\text{span}(W)$: it is a genuinely Grassmannian objective, exactly like J_{PCA} .

6.5 Gradient of the Gaussian Entropy Objective — Full Matrix Calculus

We compute $\nabla J_G(W)$ where $J_G(W) = \log \det A(W)$, $A(W) = W^\top \Sigma W$.

Step 1 — differential of $\log \det$. For a matrix-valued function A depending smoothly on W , the classical identity

$$d[\log \det A] = \text{tr}(A^{-1} dA)$$

follows from Jacobi's formula $d(\det A) = \det(A) \text{tr}(A^{-1} dA)$, divided by $\det A$.

Step 2 — differential of $A(W) = W^\top \Sigma W$. By the product rule,

$$dA = (dW)^\top \Sigma W + W^\top \Sigma (dW).$$

Step 3 — combine.

$$dJ_G = \text{tr}\left(A^{-1}[(dW)^\top \Sigma W + W^\top \Sigma dW]\right) = \text{tr}(A^{-1}(dW)^\top \Sigma W) + \text{tr}(A^{-1}W^\top \Sigma dW).$$

The first trace term, using $\text{tr}(BC) = \text{tr}(CB)$ and A^{-1} symmetric (since A is symmetric positive-definite),

$$\text{tr}(A^{-1}(dW)^\top \Sigma W) = \text{tr}(\Sigma W A^{-1}(dW)^\top) = \text{tr}((dW)^\top \Sigma W A^{-1}),$$

so both terms are transposes of one another and combine to

$$dJ_G = \text{tr}\left((dW)^\top [2 \Sigma W A^{-1}]\right).$$

By definition, the Euclidean gradient is the matrix G such that $dJ_G = \text{tr}((dW)^\top G)$, so

$$\boxed{\nabla J_G(W) = 2 \Sigma W (W^\top \Sigma W)^{-1}} \tag{6.3}$$

This is the *Euclidean* gradient, valid before any manifold constraint is imposed; §6.7 projects it onto the Stiefel tangent space.

6.6 Gradient of the Classical PCA Objective, for Comparison

For $J_{\text{PCA}}(W) = \text{tr}(W^\top \Sigma W)$,

$$dJ_{\text{PCA}} = \text{tr}((dW)^\top \Sigma W) + \text{tr}(W^\top \Sigma dW) = 2 \text{tr}((dW)^\top \Sigma W)$$

(using symmetry of Σ and the cyclic trace property as above), giving

$$\boxed{\nabla J_{\text{PCA}}(W) = 2 \Sigma W.} \tag{6.4}$$

Comparing (6.3) and (6.4): the PCA gradient is $2\Sigma W$, while the entropy gradient is $2\Sigma W (W^\top \Sigma W)^{-1}$ — the extra factor $(W^\top \Sigma W)^{-1}$ is a $k \times k$ “whitening” term coming from the log det curvature. The two gradients point in different Euclidean directions in general, which is exactly consistent with $J_G \neq J_{\text{PCA}}$ as functions — yet, as proven above, they still drive iterative algorithms toward the same stationary subspace.

6.7 Riemannian Gradient on the Stiefel Manifold

6.7.1 Tangent space

$$T_W \text{St}(d, k) = \{\Delta \in \mathbb{R}^{d \times k} : W^\top \Delta + \Delta^\top W = 0\}. \quad (6.5)$$

Derivation: differentiate the defining constraint $W(t)^\top W(t) = I_k$ along any smooth curve $W(t) \subset \text{St}(d, k)$ with $W(0) = W$, $\dot{W}(0) = \Delta$:

$$\frac{d}{dt} [W(t)^\top W(t)]_{t=0} = \Delta^\top W + W^\top \Delta = 0,$$

which is exactly the skew-symmetry-type condition defining (6.5) (note $W^\top \Delta$ need not itself be skew-symmetric, but its symmetric part must vanish).

6.7.2 Orthogonal projection of the Euclidean gradient

Given Euclidean gradient $G = \nabla J(W)$, the Riemannian gradient is its orthogonal projection onto $T_W \text{St}(d, k)$:

$$\boxed{\text{grad } J(W) = G - W \text{sym}(W^\top G)}, \quad \text{sym}(A) := \frac{1}{2}(A + A^\top). \quad (6.6)$$

Verification that (6.6) lies in the tangent space: let $\Delta = G - W \text{sym}(W^\top G)$. Then

$$\begin{aligned} W^\top \Delta + \Delta^\top W &= W^\top G - \text{sym}(W^\top G) + G^\top W - \text{sym}(W^\top G)^\top \\ &= [W^\top G - \text{sym}(W^\top G)] + [(W^\top G)^\top - \text{sym}(W^\top G)] \\ &= \text{skew}(W^\top G) - \text{skew}(W^\top G) = 0, \end{aligned}$$

using $W^\top G - \text{sym}(W^\top G) = \text{skew}(W^\top G)$ and its transpose being $-\text{skew}(W^\top G)$; the two skew terms cancel, confirming $\Delta \in T_W \text{St}(d, k)$ as required.

For the Gaussian entropy objective, substituting (6.3) into (6.6) gives the exact search direction a manifold optimizer would use at each iteration; the PCA case is identical with (6.4) in place of (6.3).

CHAPTER 7 — Manifold Geometry: Stiefel and Grassmann

7.1 The Stiefel Manifold

Definition 7.1.

$$\text{St}(d, k) = \{W \in \mathbb{R}^{d \times k} : W^\top W = I_k\}. \quad (7.1)$$

This is the set of all $d \times k$ matrices with orthonormal columns — i.e. all ordered orthonormal k -frames in $\mathbb{R}^d$. It is a compact, smooth manifold of dimension $dk - \frac{1}{2}k(k+1)$ (the ambient dk parameters minus the $\frac{1}{2}k(k+1)$ independent constraints in $W^\top W = I_k$, since that constraint is itself a symmetric $k \times k$ matrix equation).

7.1.1 Why compactness matters

Proposition 7.2. *If $J : \text{St}(d, k) \rightarrow \mathbb{R}$ is continuous, then J attains a maximum and a minimum on $\text{St}(d, k)$.*

Proof. $\text{St}(d, k)$ is a closed and bounded (hence compact, by Heine–Borel) subset of $\mathbb{R}^{d \times k}$: closed because it is the preimage of $\{I_k\}$ under the continuous map $W \mapsto W^\top W$, and bounded because $\|W\|_F^2 = \text{tr}(W^\top W) = \text{tr}(I_k) = k$ for every $W \in \text{St}(d, k)$. The extreme value theorem then guarantees a continuous function on a compact set attains its extrema. $\square$

This is the first existence guarantee for *every* objective studied in this report, including the non-smooth Laplace case (Chapter 8), as long as continuity holds.

7.2 The Grassmann Manifold

Definition 7.3. The Grassmann manifold $\text{Gr}(k, d)$ is the set of all k -dimensional linear subspaces of $\mathbb{R}^d$.

There is a natural surjection $\text{St}(d, k) \rightarrow \text{Gr}(k, d)$, $W \mapsto \text{span}(W)$, whose fibers are the orbits $\{WQ : Q \in O(k)\}$ of the orthogonal group action. Whenever an objective satisfies $J(WQ) = J(W)$ for all $Q \in O(k)$ — as proven for both J_{PCA} and J_G in Chapters 4 and 6 — the function J *descends* to a well-defined function on $\text{Gr}(k, d)$, and the true decision variable is the subspace, not a particular orthonormal basis for it.

7.2.1 Practical consequence

Optimizing on $\text{Gr}(k, d)$ instead of $\text{St}(d, k)$ when the objective is basis-invariant:

- removes the redundant $O(k)$ orbit direction from the search space, so the optimizer never wastes iterations rotating within an already-optimal subspace;

- explains, without contradiction, why the maximizer in Chapter 6 is only determined *up to rotation* — uniqueness holds at the level of the subspace (point in $\text{Gr}(k, d)$), not the specific basis (point in $\text{St}(d, k)$);
- is the natural setting whenever J is built purely from $W^\top \Sigma W$, $\det(W^\top \Sigma W)$, or other basis-free functionals of the projected second-moment structure.

If instead J depends on the individual coordinates of $Y = W^\top X$ (as coordinatewise ℓ_1 -type objectives can — see §8.2), invariance can fail, and the correct domain reverts to $\text{St}(d, k)$.

7.3 Tangent Spaces and Retractions

The tangent space $T_W \text{St}(d, k)$ was derived in (6.5). Because $\text{St}(d, k)$ is curved, a step $W + \eta \text{grad } J(W)$ generally leaves the manifold, so an update must be mapped back via a *retraction* $R_W : T_W \text{St}(d, k) \rightarrow \text{St}(d, k)$.

7.3.1 QR retraction

$$R_W(\Delta) = \text{qf}(W + \Delta), \quad (7.2)$$

where $\text{qf}(\cdot)$ denotes the Q -factor of a thin QR decomposition with positive diagonal R . *Justification:* for any full-rank $d \times k$ matrix $M = W + \Delta$, the decomposition $M = QR$ with $Q \in \text{St}(d, k)$, R upper-triangular with positive diagonal, is unique; Q is the closest orthonormal frame to M in this parametrization and reduces to W itself when $\Delta = 0$ (since $W = W \cdot I_k$ is already a valid QR factorization of W).

7.3.2 Polar retraction

$$R_W(\Delta) = (W + \Delta) [(W + \Delta)^\top (W + \Delta)]^{-1/2}. \quad (7.3)$$

Justification: this is the orthonormal matrix nearest to $W + \Delta$ in Frobenius norm (the solution of the orthogonal Procrustes problem $\min_{Q^\top Q = I_k} \|Q - (W + \Delta)\|_F$), obtained from the polar decomposition $M = Q(M^\top M)^{1/2}$, solved for Q .

Both retractions satisfy the two properties required for first-order convergence theory: $R_W(0) = W$, and $\frac{d}{dt} R_W(t\Delta) \big|_{t=0} = \Delta$ (the retraction agrees with the manifold to first order).

CHAPTER 8 Non-Gaussian Extensions

Having established the Gaussian baseline completely, we now change the distributional assumption and show, by explicit computation, how the *geometry* of the resulting objective changes.

8.1 Student- t : A Smooth, Non-Convex, Robust Objective

8.1.1 Why Student- t ?

The multivariate Student- t distribution has heavier tails than the Gaussian, making it a natural model for data contaminated by outliers or impulsive noise, and hence a natural basis for a *robust* PCA-like criterion.

8.1.2 Density and log-likelihood

For $y \in \mathbb{R}^k$, scatter matrix $S \succ 0$, degrees of freedom $\nu > 0$,

$$p(y) \propto |S|^{-1/2} \left(1 + \frac{1}{\nu} y^\top S^{-1} y \right)^{-(\nu+k)/2}. \quad (8.1)$$

Taking the log of (8.1) for a sample $y_i = W^\top x_i$ and summing over n observations gives (up to an additive constant independent of W, S) the negative log-likelihood

$$-\ell(W, S) = \frac{n}{2} \log |S| + \frac{\nu + k}{2} \sum_{i=1}^n \log \left(1 + \frac{1}{\nu} y_i^\top S^{-1} y_i \right). \quad (8.2)$$

Maximizing the likelihood is equivalent to minimizing (8.2), giving the projection objective (with S_W denoting the scatter estimated from the projected sample)

$$\boxed{J_T(W) = -\frac{n}{2} \log |S_W| - \frac{\nu + k}{2} \sum_{i=1}^n \log \left(1 + \frac{1}{\nu} y_i^\top S_W^{-1} y_i \right), \quad y_i = W^\top x_i.} \quad (8.3)$$

A simplified variant often used in practice fixes $S_W = I_k$ (or absorbs it into a normalization) and studies

$$J'_T(W) = -\sum_{i=1}^n \log \left(1 + \frac{1}{\nu} \|W^\top x_i\|^2 \right), \quad (8.4)$$

which isolates the essential non-quadratic penalty structure.

8.1.3 Gradient of the simplified Student- t objective

Write $u_i(W) = 1 + \frac{1}{\nu} \|W^\top x_i\|^2 = 1 + \frac{1}{\nu} x_i^\top W W^\top x_i$. By the chain rule,

$$d[-\log u_i] = -\frac{1}{u_i} du_i, \quad du_i = \frac{2}{\nu} x_i^\top (dW) W^\top x_i = \frac{2}{\nu} \text{tr}((dW)^\top x_i x_i^\top W),$$

using the trace/cyclic identity $x_i^\top (dW) W^\top x_i = \text{tr}(W^\top x_i x_i^\top (dW)) = \text{tr}((dW)^\top x_i x_i^\top W)$. Summing over i with the outer $-\log$,

$$dJ'_T = -\sum_{i=1}^n \frac{1}{u_i(W)} \cdot \frac{2}{\nu} \text{tr}((dW)^\top x_i x_i^\top W) = \text{tr}\left((dW)^\top \left[-\frac{2}{\nu} \sum_{i=1}^n \frac{x_i x_i^\top}{u_i(W)} W\right]\right).$$

$$\boxed{\nabla J'_T(W) = -\frac{2}{\nu} \sum_{i=1}^n \frac{1}{1 + \frac{1}{\nu} \|W^\top x_i\|^2} x_i x_i^\top W.} \quad (8.5)$$

Compare this to the PCA gradient $2\Sigma W = \frac{2}{n} \sum_i x_i x_i^\top W$ (sample version): the Student- t gradient re-weights each outer product $x_i x_i^\top$ by a factor $1/(1 + \frac{1}{\nu} \|W^\top x_i\|^2)$ that *shrinks the influence of points with large projected norm* — this is exactly the outlier down-weighting that makes the estimator robust, and it is visible directly in the gradient formula, not merely asserted.

8.1.4 Geometry of the Student- t objective

- **Continuity/smoothness:** as long as $S_W \succ 0$ and the arguments of $\log(1 + \cdot)$ stay positive (automatic, since the term inside is $1 +$ nonnegative quantity), (8.3)–(8.4) are C^∞ — the reweighting factor is a smooth function of W .
- **Non-convexity:** unlike J_{PCA} or J_G , the per-sample terms $-\log(1 + \frac{1}{\nu} \|W^\top x_i\|^2)$ are neither convex nor concave jointly with the orthogonality constraint in general position, and (8.3)–(8.4) admit no closed-form eigen-solution: the weights $1/u_i(W)$ depend on W itself, so the stationarity condition $\text{grad } J'_T(W) = 0$ is an implicit, non-linear eigenvalue-type equation rather than a plain eigenvector equation.
- **Basis invariance:** (8.4) depends on W only through $\|W^\top x_i\|^2 = x_i^\top W W^\top x_i$, and $WQ(WQ)^\top = WQ Q^\top W^\top = W W^\top$ for $Q \in O(k)$, so $J'_T(WQ) = J'_T(W)$: the objective is Grassmannian, exactly like the Gaussian case.

8.1.5 Implied optimization method

Because J_T is smooth, non-convex, and basis-invariant, the correct algorithm family is **Riemannian gradient ascent**, conjugate gradient, or trust-region methods on $\text{Gr}(k, d)$ (or $\text{St}(d, k)$), using (8.5) projected via (6.6) at every step; unlike the Gaussian case there is no eigendecomposition shortcut, and multiple random initializations are advisable because non-convexity permits spurious local maxima.

8.2 Laplace / ℓ_1 -Type: A Non-Smooth Robust Objective

8.2.1 Density and negative log-likelihood

The scalar Laplace density centered at 0 with scale b is

$$p(y) = \frac{1}{2b} \exp\left(-\frac{|y|}{b}\right) \implies -\log p(y) = \log(2b) + \frac{|y|}{b}.$$

Modeling each projected coordinate as Laplace-distributed, the negative log-likelihood of a sample is proportional to $\sum_i |y_i|$, i.e. an ℓ_1 (absolute-deviation) penalty rather than the ℓ_2 (squared) penalty implicit in the Gaussian case.

8.2.2 Candidate projection objective

$$J_L(W) = - \sum_{i=1}^n \|W^\top x_i\|_1 \quad (8.6)$$

(maximizing J_L is equivalent to minimizing total absolute deviation, mirroring how PCA minimizes total squared deviation).

8.2.3 Non-smoothness — where and why

The scalar map $z \mapsto |z|$ is not differentiable at $z = 0$; hence whenever a projected coordinate $w_j^\top x_i = 0$ exactly, the corresponding term in (8.6) is a *kink*. Its one-sided derivatives are $+1$ ($z > 0$) and -1 ($z < 0$), and the **subdifferential** at $z = 0$ is the entire interval $[-1, 1]$:

$$\partial|z| = \begin{cases} \{\text{sign}(z)\}, & z \neq 0, \\ [-1, 1], & z = 0. \end{cases} \quad (8.7)$$

This is why gradient-based analysis (as used for J_G and J_T) cannot be applied directly to (8.6) — the standard chain rule requires differentiability at the point of evaluation.

8.2.4 Invariance is no longer automatic

Proposition 8.1. $J_L(WQ) \neq J_L(W)$ in general for $Q \in O(k)$; the coordinatewise ℓ_1 norm is **not** rotationally invariant.

Sketch. Take $k = 2$, a single data point x with $W^\top x = (1, 1)^\top$, so $\|W^\top x\|_1 = 2$. Rotating by $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \in O(2)$ gives $(WQ)^\top x = Q^\top W^\top x = \frac{1}{\sqrt{2}}(0, 2)^\top$, so $\|(WQ)^\top x\|_1 = \sqrt{2} \neq 2$. Hence J_L changes under an orthogonal change of basis within the subspace: the objective depends on the specific basis, not merely on $\text{span}(W)$. $\square$

This is a genuine design fork: if a *subspace* method is desired, one should replace (8.6) with a rotationally invariant robust functional of the projected scatter (e.g. built from a robust covariance estimate of Y); if the coordinatewise ℓ_1 form is kept, the correct domain is $\text{St}(d, k)$, not $\text{Gr}(k, d)$, and the specific orthonormal basis is meaningful, not just the subspace it spans.

8.2.5 Handling non-smoothness: three standard remedies

1. **Smoothing.** Replace $|z|$ by the Huber-like approximation $\sqrt{z^2 + \varepsilon}$ for small $\varepsilon > 0$. Its derivative,

$$\frac{d}{dz} \sqrt{z^2 + \varepsilon} = \frac{z}{\sqrt{z^2 + \varepsilon}},$$

is smooth everywhere and converges to $\text{sign}(z)$ as $\varepsilon \rightarrow 0^+$, recovering the subgradient (8.7) in the limit while permitting ordinary Riemannian gradient methods at any fixed $\varepsilon > 0$.

2. **Riemannian subgradient methods.** Use any element of (8.7) at kinks and follow a diminishing-step-size subgradient scheme, projected via (6.6); convergence guarantees are weaker (typically to a neighborhood of a stationary point, at a slower $O(1/\sqrt{t})$ rate) than for smooth objectives.
3. **Majorization–minimization (MM) / iteratively reweighted least squares.** Upper-bound $|z|$ near a current iterate $z_0 \neq 0$ by the quadratic majorizer $\frac{z^2}{2|z_0|} + \frac{|z_0|}{2}$ (tangent to $|z|$ at z_0 and everywhere above it, by the AM–GM-type inequality $\frac{z^2}{2|z_0|} + \frac{|z_0|}{2} \geq |z|$), reducing each iteration to a weighted quadratic (PCA-like, closed-form) problem whose weights are updated from the previous iterate.

8.2.6 Implied optimization method

Because J_L is continuous but non-smooth, and generally not basis-invariant, the appropriate solvers are **smoothing + Riemannian gradient ascent** (simplest, recommended first choice), **Riemannian subgradient methods**, or **MM / iteratively reweighted least squares** on the Stiefel manifold $\text{St}(d, k)$ — never a direct closed-form eigendecomposition, and never (without modification) a Grassmann-only formulation.

CHAPTER 9 — A General Geometric-Analysis Checklist

Chapters 6 and 8 analyzed three specific objectives. This chapter extracts the reusable checklist: for *any* distribution-induced objective

$$J(W) = \mathbb{E}[\phi(W^\top X)] \quad \text{or} \quad J(W) = \mathcal{F}(W^\top x_1, \dots, W^\top x_n),$$

the following six questions determine which numerical method is justified.

9.1 (A) Domain: Stiefel or Grassmann?

Test whether $J(WQ) = J(W)$ for all $Q \in O(k)$ (Propositions in Chapters 4, 6, 8). If yes, optimize on $\text{Gr}(k, d)$; if no, remain on $\text{St}(d, k)$ and treat the specific basis as meaningful.

9.2 (B) Continuity

If J is continuous and $\text{St}(d, k)$ is compact, a global maximizer exists (proved generally in Chapter 7). This holds for all three case studies: J_{PCA} , J_G , J_T , and J_L are all continuous on the feasible set (the Student- t and Gaussian cases additionally require the relevant scatter matrix to remain positive definite, which holds generically).

9.3 (C) Differentiability

Objective	Differentiable?	Where derived
$J_{\text{PCA}} = \text{tr}(W^\top \Sigma W)$	Yes, C^∞ (quadratic)	(6.4)
$J_G = \log \det(W^\top \Sigma W)$	Yes, C^∞	(6.3)
J'_T (Student- t)	Yes, C^∞	(8.5)
J_L (Laplace, ℓ_1)	No — kinks at zero crossings	(8.7)

9.4 (D) Gradient / Hessian

Explicit first-order derivatives were obtained for every smooth case ((6.4), (6.3), (8.5)); a subdifferential was obtained for the non-smooth case ((8.7)). Second-order information (Riemannian Hessians) is not required for first-order methods but becomes relevant if trust-region or Newton-type solvers are used, particularly for the non-convex J_T , where curvature helps distinguish saddle points from maxima.

9.5 (E) Landscape: Convexity and Critical Points

-
- J_{PCA}, J_G : closed-form global optimum (top- k eigenspace); all other coordinate-subspace stationary points are saddles or local optima ordered by which eigenvalues are selected (Theorem 4.2 and its log det analogue).
 - J_T : generally non-convex with no closed form; local maxima depend on the outlier structure of the data and the weight function $1/u_i(W)$; multiple initializations are needed to guard against spurious local optima.
 - J_L : non-smooth and (in the coordinatewise form) not basis-invariant; the landscape includes non-differentiable ridges wherever a projected coordinate crosses zero.

9.6 (F) Existence and Uniqueness

Existence of a maximizer follows from compactness plus continuity (§B) in all four cases. Uniqueness holds only up to the residual symmetry of the problem: up to $O(k)$ rotation for the Grassmannian objectives (J_{PCA}, J_G, J_T), and, for J_L , potentially unique at the basis level, though possibly with multiple disconnected optima due to non-convexity and non-smoothness.

9.7 (G) Numerical Implication — the Master Decision Rule

The preceding five questions collapse into one operational rule, tabulated fully as Chapter 11's decision table: smooth + closed form $\Rightarrow$ eigendecomposition; smooth + non-convex $\Rightarrow$ Riemannian gradient/trust-region; non-smooth $\Rightarrow$ smoothing, subgradient, or MM.

CHAPTER 10 — Optimization Algorithms for Manifold-Constrained Objectives

10.1 Riemannian Gradient Ascent

Given a smooth objective J and step size $\eta_t > 0$, the basic update is

$$W_{t+1} = R_{W_t}(\eta_t \operatorname{grad} J(W_t)), \quad (10.1)$$

with $\operatorname{grad} J$ from (6.6) and R_W a retraction ((7.2) or (7.3)). **Stationarity criterion:** iterate until $\|\operatorname{grad} J(W_t)\|_F < \text{tol}$. This is the workhorse method for J_G and J_T (Chapters 6 and 8), and — with a smoothed absolute value — also usable for J_L .

10.2 Riemannian Conjugate Gradient and Trust-Region Methods

When convergence of plain gradient ascent is slow (common near flat regions of J_T , where the reweighting factors in (8.5) vary slowly), conjugate-gradient directions $\xi_t = \operatorname{grad} J(W_t) + \beta_t \mathcal{T}_{W_{t-1} \rightarrow W_t}(\xi_{t-1})$ (with $\mathcal{T}$ a vector transport and β_t a Riemannian analogue of the Fletcher–Reeves or Polak–Ribière coefficient) typically accelerate convergence. Trust-region methods additionally use curvature (the Riemannian Hessian) to choose step length and are preferred when reliable escape from saddle points is required for the non-convex J_T landscape.

10.3 Subgradient, Smoothing, and MM Methods for Non-Smooth Objectives

For J_L (§8.2):

- **Smoothing** solves a sequence of smooth problems with $\varepsilon_t \rightarrow 0$, each solved by (10.1), warm-started from the previous solution — the recommended default.
- **Riemannian subgradient** replaces $\operatorname{grad} J$ in (10.1) with any Riemannian subgradient built from (8.7), using a diminishing step size $\eta_t = c/\sqrt{t}$ for convergence in the (weaker) subgradient sense.
- **MM / iteratively reweighted least squares** solves, at each outer iteration, a *weighted* PCA-type problem (closed-form via eigendecomposition of a reweighted scatter matrix), using the majorizer of §8.2; this is often the fastest and most numerically stable option because each subproblem reuses the exact closed-form machinery of Chapter 4.

10.4 Stopping Criteria and Practical Diagnostics

Across all methods, track: (i) objective value $J(W_t)$ versus iteration (should be monotonically non-decreasing for exact gradient ascent with sufficiently small step size, by a standard sufficient-decrease/Armijo-type argument); (ii) Riemannian gradient norm $\|\text{grad } J(W_t)\|_F$ versus iteration (should approach 0 for smooth objectives); (iii) principal angles between $\text{span}(W_t)$ and the previous iterate's subspace, to detect convergence of the subspace even if the specific basis continues to rotate.

CHAPTER 11 —Method Selection: The Decision Table

Table 11.1 consolidates every geometric property derived in Chapters 6, 8, and 9 into a single lookup table connecting an objective’s mathematical properties to a justified optimization choice.

Objective property	Meaning	Justified optimization choice
Closed-form spectral objective	J_{PCA}, J_G : reduces to an eigenvalue problem (Ch. 4, 6)	Direct eigendecomposition; manifold method optional, for validation only
Smooth orthogonality-constrained	+ Gradient exists on the manifold (all of Ch. 6, 8 except J_L)	Riemannian gradient ascent (10.1)
Smooth + curvature matters	Slow convergence / ill-conditioning (e.g. J_T near flat regions)	Riemannian conjugate gradient or trust-region
Smooth invariant + basis-	True variable is a subspace (J_{PCA}, J_G, J_T)	Optimize on $\text{Gr}(k, d)$
Smooth, non-convex, robust	Student- t objective J_T (Ch. 8)	Riemannian gradient / conjugate gradient / trust-region, multiple restarts
Non-smooth objective	Laplace / ℓ_1 objective J_L (Ch. 8)	Smoothing, Riemannian subgradient, or MM
Sample-noise / estimated objective	e.g. kernel-density entropy estimates	Stochastic or mini-batch Riemannian optimization
Not rotationally invariant	Basis matters, not only subspace (J_L in coordinatewise form)	Optimize on $\text{St}(d, k)$, not $\text{Gr}(k, d)$

Table 11.1: Master decision table linking objective geometry (derived in Chapters 4, 6, 8) to a justified numerical method.

11.1 Reading the Table as a Decision Procedure

In practice, apply the checklist of Chapter 9 to a new candidate objective in this order: (1) check differentiability — if non-smooth, proceed via smoothing/subgradient/MM regardless of other properties; (2) if smooth, check for a closed-form spectral reduction — if present, prefer eigendecomposition for speed and exactness; (3) otherwise check basis invariance to decide between $\text{St}(d, k)$ and $\text{Gr}(k, d)$; (4) check convexity/landscape complexity to decide between plain gradient ascent and trust-region/multi-restart strategies.

CHAPTER 12 Experimental Design

12.1 Synthetic Data Generation with Known Ground Truth

To evaluate subspace recovery quantitatively, generate data with a known latent subspace $U \in \text{St}(d, k)$:

$$z \sim p_{\text{latent}}(z) \text{ on } \mathbb{R}^k, \quad x = Uz + \varepsilon, \quad \varepsilon \sim p_{\text{noise}}, \quad (12.1)$$

where p_{latent} and p_{noise} are chosen to match each case study:

1. **Gaussian case:** $z \sim \mathcal{N}(0, \Lambda)$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_d)$.
2. **Student- t case:** z drawn from a multivariate Student- t with small ν (heavy tails), or Gaussian z contaminated with a small fraction of large-norm outliers.
3. **Laplace case:** z drawn coordinatewise from independent Laplace distributions, or Gaussian data with sparse, large-magnitude outliers added directly to x .

12.2 Methods to Compare

- Classical PCA ((4.2), closed form).
- Gaussian entropy objective J_G ((6.2), Riemannian gradient ascent, validated against the closed form).
- Student- t objective J_T ((8.3)/(8.4), Riemannian gradient/trust-region).
- Laplace / smoothed- ℓ_1 objective J_L ((8.6), smoothing + Riemannian gradient ascent).

12.3 Evaluation Metrics

12.3.1 Subspace recovery error via principal angles

Given the estimated subspace $\widehat{W}$ and the true generating subspace U , compute the singular values $\sigma_1 \geq \dots \geq \sigma_k$ of $U^\top \widehat{W}$; the principal angles are $\theta_j = \arccos(\sigma_j)$, and a scalar recovery error is

$$d(\widehat{W}, U) = \left(\sum_{j=1}^k \sin^2 \theta_j \right)^{1/2}, \quad (12.2)$$

which is exactly 0 when the subspaces coincide and is invariant to the choice of orthonormal basis within each subspace — appropriate for comparing Grassmannian objectives, and computable regardless of whether the estimating objective was itself basis-invariant.

12.3.2 Additional metrics

1. Reconstruction error $\|X - \widehat{W}\widehat{W}^\top X\|^2$.

2. Explained variance ratio $\text{tr}(\widehat{W}^\top \Sigma \widehat{W}) / \text{tr}(\Sigma)$.
3. Robustness: recovery error (12.2) as a function of the fraction of contaminating outliers.
4. Optimization diagnostics: $J(W_t)$ and $\|\text{grad } J(W_t)\|_F$ versus iteration t (Chapter 10, §*Stopping Criteria*).
5. Sensitivity to initialization: variance of the final recovery error (12.2) across random restarts, particularly diagnostic for the non-convex J_T .

12.4 Recommended Plots

1. Objective value vs. iteration, one curve per method.
2. Gradient norm vs. iteration (log scale), one curve per method.
3. Reconstruction error vs. outlier fraction, comparing PCA, J_T , and J_L .
4. Subspace recovery error (12.2) vs. sample size n , for all methods.
5. A 2-D toy visualization: project a small synthetic dataset with outliers using each of the four objectives side by side, so the robustness difference is directly visible.
6. Wall-clock runtime comparison, since closed-form PCA is expected to be orders of magnitude faster than the iterative manifold methods.

CHAPTER 13 _____ Implementation Roadmap

13.1 Phase 1 — PCA Baseline

1. Generate synthetic Gaussian data via (12.1).
2. Implement PCA by eigendecomposition of the sample covariance.
3. Implement the identical objective (4.2) via Riemannian gradient ascent (10.1), using (6.4) and (6.6), purely as a sanity check.
4. Verify both recover the same subspace, using $(12.2) \approx 0$.

13.2 Phase 2 — Gaussian Entropy Objective

1. Implement $J_G(W) = \log \det(W^\top \Sigma W)$ ((6.2)).
2. Implement its gradient (6.3) and Riemannian projection (6.6).
3. Optimize with Riemannian gradient ascent (10.1).
4. Verify convergence to the same subspace as Phase 1 (confirming Proposition in §6.3 empirically).

13.3 Phase 3 — Student- t Objective

1. Implement the simplified objective (8.4) (or the full form (8.3) with an alternating scatter update).
2. Implement gradient (8.5).
3. Optimize with Riemannian gradient ascent or trust-region; run multiple random initializations to probe non-convexity.
4. Compare recovery under outlier contamination against PCA.

13.4 Phase 4 — Laplace / ℓ_1 -Style Objective

1. Start with the smoothed objective using $\sqrt{z^2 + \varepsilon}$ in place of $|z|$, with ε annealed toward 0.
2. Optimize on $\text{St}(d, k)$ using (10.1) at each ε -level, warm-started from the previous solution.
3. Compare against the MM/iteratively-reweighted-least-squares variant described in §8.2.
4. Evaluate robustness and convergence behavior, noting the expected slower/rougher convergence relative to the smooth cases.

CHAPTER 14 Conclusion and Future Directions

14.1 What Was Established

This report developed a single governing optimization problem, $W^* = \arg \max_{W^\top W = I_k} J(W)$, and showed concretely, with full derivations rather than assertions, that:

1. classical PCA is recovered exactly as the special case $J = J_{\text{PCA}} = \text{tr}(W^\top \Sigma W)$ (Chapter 4);
2. under a Gaussian assumption, maximizing differential entropy gives a genuinely different objective, $J_G = \log \det(W^\top \Sigma W)$, that nonetheless shares PCA's maximizer — a theorem, proved via the shared order-preserving structure of $\lambda \mapsto \lambda$ and $\lambda \mapsto \log \lambda$ (Chapter 6);
3. departing from Gaussianity changes the *geometry*, not just the numerical value, of the objective: Student- t likelihood gives a smooth but non-convex, closed-form-free objective (§8.1), and Laplace likelihood gives a non-smooth, in general non-basis-invariant objective (§8.2);
4. these geometric properties — smoothness, convexity, basis invariance, closed-formness — determine, rather than merely suggest, a specific optimization method for each case (Chapters 9–11).

14.2 Limitations

The Student- t and Laplace objectives studied here are representative but not exhaustive; other heavy-tailed or skewed families (e.g. generalized Gaussian, skew-normal, mixture models) would populate the same checklist differently. The mutual-information objective (§3.2.3) was flagged as requiring regularization or a noisy-channel model rather than being directly optimizable in the noiseless linear setting, and was not developed into a full case study here. Finally, all convergence statements for the non-convex and non-smooth cases are *local* (stationarity), not global-optimality guarantees.

14.3 Future Directions

- Extend the checklist to mixture-model and multimodal distributional assumptions, where $J(W)$ may have genuinely disconnected optima even in the Grassmannian, basis-invariant case.
- Develop a regularized, estimable surrogate for the mutual information objective (3.4) (e.g. via a noisy channel or a variational lower bound) as a fourth full case study.
- Extend the Riemannian trust-region analysis of Chapter 10 with explicit Riemannian Hessian derivations for J_T , to formally characterize which stationary points are strict saddles.
- Carry out the full empirical protocol of Chapter 12 and report the resulting plots as a companion to this mathematical report.

CHAPTER A — Consolidated Reference Tables

A.1 All Objectives and Gradients Derived in This Report

Model	Objective $J(W)$	Gradient / subgradient
PCA (variance)	$\text{tr}(W^\top \Sigma W)$ (4.2)	$2\Sigma W$ (6.4)
Gaussian en-tropy	$\log \det(W^\top \Sigma W)$ (6.2)	$2\Sigma W(W^\top \Sigma W)^{-1}$ (6.3)
Student- t	$-\sum_i \log(1 + \frac{1}{\nu} \ W^\top x_i\ ^2)$ (8.4)	$-\frac{2}{\nu} \sum_i \frac{x_i x_i^\top}{1 + \frac{1}{\nu} \ W^\top x_i\ ^2} W$ (8.5)
Laplace (ℓ_1)	$-\sum_i \ W^\top x_i\ _1$ (8.6)	subdifferential (8.7), or smoothed gradient $z/\sqrt{z^2 + \varepsilon}$ per coordinate

A.2 Notation Summary

Symbol	Meaning
$X \in \mathbb{R}^d$	Original (ambient) random data vector
$Y = W^\top X \in \mathbb{R}^k$	Reduced / projected representation
$W \in \mathbb{R}^{d \times k}$	Projection matrix, $W^\top W = I_k$
$\Sigma = \mathbb{E}[XX^\top]$	Covariance matrix of X (mean zero)
$h(\cdot)$	Differential entropy
$H(\cdot)$	Shannon (discrete) entropy
$I(\cdot; \cdot)$	Mutual information
$\text{St}(d, k)$	Stiefel manifold, $\{W : W^\top W = I_k\}$
$\text{Gr}(k, d)$	Grassmann manifold of k -dim subspaces of $\mathbb{R}^d$
$\text{grad } J(W)$	Riemannian gradient of J at W
$\nabla J(W)$	Euclidean gradient of J at W
$R_W(\cdot)$	Retraction from the tangent space back to the manifold
ν	Degrees of freedom of the Student- t distribution
$\text{sym}(A)$	Symmetric part $\frac{1}{2}(A + A^\top)$ of a matrix A